

Deciding whether the fractional dichromatic number of a digraph is at most 2 is NP-complete

github.com/graph-theory-AI

Machine-generated note (see provenance box)

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Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The reduction and proof were found by the language model gpt-5.6-sol in a single autonomous pass on 2026-09-01, as part of a large sweep over open problems catalogued from arXiv (catalog id 1812.02420__02; original writeup in attacks/1812.02420__02/output.md). (2) An independent adversarial referee agent (claude-fable-5, 2026-09-02) re-derived every step, downloaded the source paper's $\text{\LaTeX}$ source to confirm the problem statement, the LP definition and the digon convention, confirmed the external hardness result of Bokal et al., and verified the reduction by exact LP computation on 125 instances (including the digon-free Paley tournament ST_7 and the bidirected K_3); its verdict was CONFIRMED. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-09 the writeup was rewritten into the present note by claude-fable-5.1: the text was reorganised with full proofs in [Appendix A](#), and three clarifications supplied by the referee's report were added as remarks (the empty instance, attainment of the LP optimum, and the fact that the source paper itself proves NP membership). No mathematical content was changed. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

Abstract

The fractional dichromatic number $\bar{\chi}_f(D)$ of a digraph D is the optimum of the linear programme covering the vertices by acyclic vertex sets. Hochstättler, Schröder and Steiner proved that deciding $\bar{\chi}_f(D) \leq p$ is NP-complete for every real $p > 1$ with $p \neq 2$, and left the case $p = 2$ open (Problem 3.21). We show that the case $p = 2$ is NP-complete as well, for digraphs in which antiparallel arcs (digons) are allowed, as in the source paper. The reduction is from the NP-complete problem of deciding whether a digraph is 2-dicolourable (Bokal, Fijavž, Juvan, Kayll and Mohar): take two disjoint copies of D and join the two copies of each vertex by a digon. A fractional colouring of weight 2 is then forced to use only acyclic sets containing exactly one copy of every vertex, and such a set is precisely an acyclic 2-colouring of D .

1 Introduction

We follow the conventions of Hochstättler, Schröder and Steiner [2]: digraphs are finite and loopless, but parallel and antiparallel arcs are allowed, and a pair of antiparallel arcs is a directed cycle of length two, a *digon*. A set $A \subseteq V(D)$ is *acyclic* if the induced subdigraph $D[A]$ has no directed cycle; $\mathcal{A}(D)$ denotes the family of acyclic vertex sets. The *dichromatic number* $\bar{\chi}(D)$ is the least k such that $V(D)$ can be partitioned into k acyclic sets, and the *fractional dichromatic number* $\bar{\chi}_f(D)$ is the optimum of the linear programme

$$\begin{aligned} \min \quad & \sum_{A \in \mathcal{A}(D)} x_A \\ \text{subject to} \quad & \sum_{A \in \mathcal{A}(D), v \in A} x_A \geq 1 \quad \text{for every } v \in V(D), \\ & x_A \geq 0 \quad \text{for every } A \in \mathcal{A}(D). \end{aligned} \tag{1}$$

A feasible solution of (1) is a *fractional acyclic colouring* and its objective value its *weight*; since (1) is a finite feasible linear programme bounded below, its optimum is attained, so $\bar{\chi}_f(D) \leq p$ if and only if some fractional acyclic colouring has weight at most p .

Bokal, Fijavž, Juvan, Kayll and Mohar [1] proved that deciding $\bar{\chi}(D) \leq 2$ is NP-complete. Hochstättler, Schröder and Steiner studied the analogous threshold problems for fractional parameters:

Problem 1.1 ([2, Problem 3.21]). Let $p \geq 1$ be a fixed real number. Instance: a (multi-)digraph D . Decide whether $\bar{\chi}_f(D) \leq p$.

They proved that this problem is in NP for every $p \geq 1$ [2, Observation 3.15] and NP-complete for every real $p > 1$ with $p \neq 2$ [2, Theorem 3.20]: for $p > 2$ by a reduction from the fractional chromatic number of undirected graphs, and for $1 < p < 2$ by a splitting operation that maps every threshold $q > 1$ into $(1, 2)$. Neither technique reaches $p = 2$, and their concluding remarks state that they “cannot rule out the possibility that there is some clever way to algorithmically decide whether a given digraph has fractional dichromatic number at most 2”, while conjecturing NP-completeness “using other techniques”. We confirm that prediction.

Let FDC(2) denote Problem 1.1 with $p = 2$.

Theorem 1.2 (Main theorem). FDC(2) is NP-complete. NP-hardness holds already for digraphs whose vertex set is arranged in two layers, each inducing a copy of the same digraph, with the two copies of every vertex joined by a digon and no other arcs between the layers.

Corollary 1.3. Deciding $\bar{\chi}_f(D) \leq p$ for (multi-)digraphs is NP-complete for every real $p > 1$.

Proof. Combine Theorem 1.2 with [2, Theorem 3.20], which covers every $p > 1$ with $p \neq 2$. $\square$

Idea of proof. Membership in NP is standard (an optimal basic solution of (1) has at most $|V|$ nonzero entries of polynomial bit size) and is in any case [2, Observation 3.15]. For hardness, given a nonempty digraph D build $R(D)$ from two disjoint copies $D \times \{0\}$, $D \times \{1\}$ of D by adding the digon between $(v, 0)$ and $(v, 1)$ for every v . An acyclic set of $R(D)$ contains at most one end of each digon, so summing the two covering constraints of a digon shows that a fractional acyclic colouring of weight at most 2 has weight exactly 2 and every set in its support meets every digon in exactly one vertex (Lemma 2.2). Such a set selects one copy of each vertex; the vertices selected in layer 0 and in layer 1 induce acyclic subdigraphs of D and partition $V(D)$, so $\bar{\chi}(D) \leq 2$. Conversely an acyclic 2-colouring (C_0, C_1) of D gives the two complementary acyclic sets $(C_0 \times \{0\}) \cup (C_1 \times \{1\})$ and $(C_0 \times \{1\}) \cup (C_1 \times \{0\})$, which with weight 1 each form a fractional colouring of weight 2 (Proposition 2.3). Hence $\bar{\chi}_f(R(D)) \leq 2$ iff $\bar{\chi}(D) \leq 2$, and [1] finishes. [full proof]

2 The reduction

Definition 2.1 (The digraph $R(D)$). Let D be a digraph. The digraph $H = R(D)$ has vertex set $V(D) \times \{0, 1\}$; write $v_i = (v, i)$. Its arcs are

- $u_i v_i$ for every arc $uv \in A(D)$ and each $i \in \{0, 1\}$ (so each layer $V(D) \times \{i\}$ induces a copy of D), and
- $v_0 v_1$ and $v_1 v_0$ for every $v \in V(D)$ (the *matching digons*).

There are no other arcs; in particular the only arcs between the two layers are the matching digons. $R(D)$ has $2|V(D)|$ vertices and $2|A(D)| + 2|V(D)|$ arcs, and is computable from D in polynomial time.

Lemma 2.2 (Forcing lemma). Let D be a nonempty digraph, $H = R(D)$, and let $(x_A)_{A \in \mathcal{A}(H)}$ be a fractional acyclic colouring of H of weight $W = \sum_A x_A \leq 2$. Then $W = 2$, and every $A \in \mathcal{A}(H)$ with $x_A > 0$ satisfies $|A \cap \{v_0, v_1\}| = 1$ for every $v \in V(D)$.

Idea of proof. No acyclic set contains both ends of a digon. Adding the covering constraints at v_0 and v_1 gives $2 \leq \sum_A x_A |A \cap \{v_0, v_1\}| \leq \sum_A x_A = W \leq 2$, so all inequalities are equalities, and the nonnegative terms $x_A(1 - |A \cap \{v_0, v_1\}|)$ must all vanish. [\[full proof\]](#)

Proposition 2.3. *For every nonempty digraph D ,*

$$\vec{\chi}_f(R(D)) \leq 2 \iff \vec{\chi}(D) \leq 2.$$

Idea of proof. $(\Rightarrow)$ Take any support set A of a fractional colouring of weight at most 2; by [Lemma 2.2](#) the sets $C_i = \{v : v_i \in A\}$ partition $V(D)$, and the copy of $D[C_i]$ in layer i is an induced subdigraph of the acyclic $H[A]$. $(\Leftarrow)$ Given an acyclic partition (C_0, C_1) , the sets $A = (C_0 \times \{0\}) \cup (C_1 \times \{1\})$ and $B = V(H) \setminus A$ each contain one end of every matching digon and induce disjoint unions of copies of $D[C_0]$ and $D[C_1]$ in different layers, hence are acyclic; weights $x_A = x_B = 1$ cover every vertex. [\[full proof\]](#)

Lemma 2.4 (Membership in NP). $\text{FDC}(2) \in \text{NP}$.

Idea of proof. The LP (1) has a 0/1 constraint matrix; an optimal basic feasible solution has at most $N = |V(H)|$ positive variables, with rational values of $O(N \log N)$ bits by Cramer’s rule and Hadamard’s bound. The certificate lists these sets and weights, and acyclicity, coverage and total weight are checked in polynomial time. (This is also [\[2, Observation 3.15\]](#).) [\[full proof\]](#)

3 Remarks

Remark 3.1 (Digons are essential). The proof uses the antiparallel pair v_0v_1, v_1v_0 in an essential way: it is what prevents an acyclic set from containing both copies of a vertex. [Theorem 1.2](#) therefore settles [Problem 1.1](#) as stated (its instances are multidigraphs, and [\[2\]](#) explicitly treats a digon as a directed 2-cycle), but it does not settle the variant in which the input is required to be an oriented graph (digon-free). That variant remains open.

Remark 3.2 (Degenerate instances). [Lemma 2.2](#) and [Proposition 2.3](#) assume $V(D) \neq \emptyset$, as the original writeup does. The empty digraph is a trivial yes-instance of both problems, so the reduction $D \mapsto R(D)$ is correct on all inputs. Since digraphs are loopless, every singleton is acyclic and (1) is always feasible.

Remark 3.3 (Why the earlier techniques stop short of $p = 2$). As the referee’s report observes ([Appendix B](#)), the hardness proof of [\[2\]](#) for $p > 2$ reduces from “ $\chi_f(G) \leq p$ ” for undirected graphs, which is polynomial at $p = 2$ (bipartiteness), while the proof for $1 < p < 2$ pushes thresholds $q > 1$ through an ℓ -split to $\ell q / ((\ell - 1)q + 1) < 2$. Neither reaches exactly 2. The present reduction instead starts from the NP-completeness of 2-dicolourability [\[1\]](#).

Remark 3.4 (Computational check). The referee verified [Proposition 2.3](#) by brute force (enumeration of all acyclic subsets of $R(D)$ and exact solution of (1)) on 125 instances: the directed triangle ($\vec{\chi}_f(R(C_3)) = 2$), the bidirected K_3 ($\vec{\chi}_f = 3$), a single digon, a single vertex, the digon-free Paley tournament ST_7 with $\vec{\chi} = 3$ ($\vec{\chi}_f(R(ST_7)) = 7/3$), and 120 random digraphs on at most 6 vertices. In every yes-instance every optimal support set met each matching digon in exactly one vertex, as [Lemma 2.2](#) predicts. Script: `verification/scripts/1812.02420__02/check_reduction.py`.

Remark 3.5 (Novelty). The referee’s literature search found no resolution of the $p = 2$ case after [\[2\]](#); the catalogue’s own review (May 2026) found none either. The argument is short, which makes an unindexed prior appearance conceivable. This has not been checked by a human.

A Full proofs

Lemma 2.2 (Forcing lemma). *Let D be a nonempty digraph, $H = R(D)$, and let $(x_A)_{A \in \mathcal{A}(H)}$ be a fractional acyclic colouring of H of weight $W = \sum_A x_A \leq 2$. Then $W = 2$, and every $A \in \mathcal{A}(H)$ with $x_A > 0$ satisfies $|A \cap \{v_0, v_1\}| = 1$ for every $v \in V(D)$.*

Proof. Fix $v \in V(D)$. The vertices v_0, v_1 form a digon in H , so no acyclic set contains both, i.e. $|A \cap \{v_0, v_1\}| \leq 1$ for every $A \in \mathcal{A}(H)$. Adding the covering constraints of (1) at v_0 and at v_1 ,

$$2 \leq \sum_{A \ni v_0} x_A + \sum_{A \ni v_1} x_A = \sum_{A \in \mathcal{A}(H)} x_A |A \cap \{v_0, v_1\}| \leq \sum_{A \in \mathcal{A}(H)} x_A = W \leq 2.$$

Hence all inequalities are equalities; in particular $W = 2$ and

$$\sum_{A \in \mathcal{A}(H)} x_A (1 - |A \cap \{v_0, v_1\}|) = 0.$$

Every summand is nonnegative, since $x_A \geq 0$ and $|A \cap \{v_0, v_1\}| \leq 1$. Therefore each summand is zero, and for every A with $x_A > 0$ we get $|A \cap \{v_0, v_1\}| = 1$. As v was arbitrary, the claim follows. $\square$

Proposition 2.3. *For every nonempty digraph D ,*

$$\vec{\chi}_f(R(D)) \leq 2 \iff \vec{\chi}(D) \leq 2.$$

Proof. Write $H = R(D)$.

Forward implication. Assume $\vec{\chi}_f(H) \leq 2$. Since the optimum of (1) is attained, there is a fractional acyclic colouring (x_A) of H of weight at most 2, and it has a support set A with $x_A > 0$ (the weight is at least 1 because D is nonempty and some vertex must be covered). By Lemma 2.2, A contains exactly one of v_0, v_1 for every $v \in V(D)$, so the sets

$$C_i = \{v \in V(D) : v_i \in A\}, \quad i \in \{0, 1\},$$

partition $V(D)$. For each i , the vertex set $C_i \times \{i\} \subseteq A$ induces in H a copy of $D[C_i]$ (layer i induces a copy of D), and this copy is an induced subdigraph of $H[A]$. Since A is acyclic, $H[A]$ has no directed cycle, hence neither do its induced subdigraphs, and $D[C_0], D[C_1]$ are acyclic. Thus (C_0, C_1) is a partition of $V(D)$ into two acyclic sets and $\vec{\chi}(D) \leq 2$.

Reverse implication. Assume $V(D) = C_0 \sqcup C_1$ with $D[C_0]$ and $D[C_1]$ acyclic. Put

$$A = (C_0 \times \{0\}) \cup (C_1 \times \{1\}), \quad B = (C_0 \times \{1\}) \cup (C_1 \times \{0\}) = V(H) \setminus A.$$

Each matching digon $\{v_0, v_1\}$ has one end in A and one in B , so neither $H[A]$ nor $H[B]$ contains a matching arc; as the matching digons are the only arcs between the layers, $H[A]$ has no arcs between $C_0 \times \{0\}$ and $C_1 \times \{1\}$. Hence

$$H[A] \cong D[C_0] \sqcup D[C_1],$$

the two terms lying in different layers, and $H[A]$ is acyclic. Exchanging the roles of the layers, $H[B] \cong D[C_1] \sqcup D[C_0]$ is acyclic as well. Setting $x_A = x_B = 1$ and $x_{A'} = 0$ for all other acyclic sets gives every vertex of H coverage exactly 1 (each vertex lies in exactly one of A, B) with total weight 2. Hence $\vec{\chi}_f(H) \leq 2$. $\square$

Lemma 2.4 (Membership in NP). $\text{FDC}(2) \in \text{NP}$.

Proof. Let H be an instance with $N = |V(H)|$ vertices. The LP (1) for H has N covering constraints, so an optimal basic feasible solution has at most N positive variables. The constraint matrix has entries in $\{0, 1\}$, so by Cramer's rule the coordinates of a basic solution are ratios of determinants of $N \times N$ 0/1-matrices, whose absolute values are at most $N^{N/2}$ by Hadamard's inequality; numerators and denominators therefore have $O(N \log N)$ bits. A certificate for a yes-instance lists at most N vertex sets together with their rational weights. The verifier checks in polynomial time that each listed set induces an acyclic subdigraph of H , that every vertex receives total weight at least 1, and that the total weight is at most 2. Hence $\text{FDC}(2) \in \text{NP}$. $\square$

Theorem 1.2 (Main theorem). $\text{FDC}(2)$ is NP-complete. NP-hardness holds already for digraphs whose vertex set is arranged in two layers, each inducing a copy of the same digraph, with the two copies of every vertex joined by a digon and no other arcs between the layers.

Proof. By Lemma 2.4, $\text{FDC}(2) \in \text{NP}$. For hardness, the map $D \mapsto R(D)$ of Definition 2.1 is computable in polynomial time, and by Proposition 2.3 (together with Remark 3.2 for the empty digraph) it maps yes-instances of “ $\bar{\chi}(D) \leq 2$ ” to yes-instances of $\text{FDC}(2)$ and no-instances to no-instances. Since deciding $\bar{\chi}(D) \leq 2$ is NP-complete [1], this is a polynomial-time many-one reduction and $\text{FDC}(2)$ is NP-hard. The digraphs $R(D)$ have exactly the two-layer structure described in the statement. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-fable-5, 2026-09-02) on the *original* writeup. Its step numbers refer to that writeup, not to this note. The scripts it mentions are in `verification/scripts/1812.02420__02/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The writeup formalizes Problem 3.21 of arXiv:1812.02420 (Hochstättler, Schröder, Steiner, DMTCS 22:1, 2020) exactly as the paper does. I downloaded the arXiv v4 LaTeX source (identical to the DMTCS version) and checked:

- **Digons are in scope.** The paper’s conventions section states verbatim: “Graphs and digraphs in this paper are considered loopless, but are allowed to have multiple parallel and anti-parallel edges/arcs between vertices ... a directed cycle of length two (a *digon*) is a pair of anti-parallel arcs in a digraph.” Problem 3.21 takes “A (multi-)digraph D ” as instance. So a reduction whose output contains digons is squarely within the intended problem — this is not interpretation gaming.
- ****The definition of χ_f^* matches.**** The paper defines $\chi_f^*(D)$ as the optimal value of the covering LP: $\min \sum_{A \in \mathcal{A}(D)} x_A$ subject to $\sum_{A \ni v} x_A \geq 1$ for all v , $x \geq 0$, where $\mathcal{A}(D)$ is the family of vertex sets inducing acyclic subdigraphs. The writeup uses exactly this LP.
- **The intended open question is NP-hardness at $p = 2$, digons allowed.** The paper’s conclusion says verbatim: “Theorem 3.20 furthermore does not treat the case $p=2$. Thus, we cannot rule out the possibility that there is some clever way to algorithmically decide whether a given digraph has fractional dichromatic number at most 2. Still, the authors strongly believe that using other techniques, it should be possible to prove NP-completeness also in this case.” A proof of NP-completeness for multidigraph inputs is exactly what the authors asked for; a reasonable author of the paper would consider the problem resolved by this argument. (The writeup itself honestly flags that the oriented/digon-free variant is *not* settled — but that variant is not Problem 3.21.)

Why the authors’ own techniques missed $p = 2$ is also visible in the source: their $p > 2$ hardness reduces from $\chi_f(G) \leq p$ for undirected graphs (NP-hard only for $p > 2$; $p = 2$ is bipartiteness, polynomial), and their $p \in (1, 2)$ hardness pushes values through the l -split operation, which maps any $q > 1$ to $lq/((l-1)q+1) < 2$. Neither technique reaches exactly 2. The writeup’s reduction from Bokal et al.’s 2-dichromatic hardness is precisely the kind of “other technique” the authors anticipated.

B.2 Step-by-step findings

step	label	note
1. Problem statement / digon convention	VALID	Matches the paper's Problem 3.21 and its explicit convention that a digon is a directed 2-cycle and anti-parallel arcs are allowed.
2. LP definition of $\chi_{\rightarrow}^f$	VALID	Identical to the paper's Definition (LP "(1)"/(primal)); the LP min is attained (finite LP, feasible when D is loopless, bounded below), so " $\chi_{\rightarrow}^f(H) \leq 2$ " is equivalent to existence of a feasible solution of weight ≤ 2 . Standard vertex-of-polyhedron argument: $\leq N$ positive variables at a basic optimum, Cramer + Hadamard give $O(N \log N)$ -bit rationals. Moreover the source paper itself proves NP membership for every $p \geq 1$ (Observation 3.15 following Problem 3.21), by essentially the same argument, so this step is even citable.
3. Membership in NP	VALID	
4. Hardness source: deciding $\chi(\vec{D}) \leq 2$ is NP-complete	VALID	Bokal, Fijavž, Juvan, Kayll, Mohar, "The circular chromatic number of a digraph", J. Graph Theory 46(3):227–240 (2004); confirmed by web search (reduction from 2-colouring 3-uniform hypergraphs) and quoted in the source paper's own abstract.
5. Construction $R(D)$ (two layers, matching digons)	VALID	Well-defined for multidigraphs, polynomial: $2 V(D) $ vertices, $2 A(D) +2 V(D) $ arcs, as claimed. $R(D)$ is the Cartesian product $D \square K \leftrightarrow_2$.
6. Forcing lemma	VALID	Re-derived: for each v , the two coverage constraints give $2 \leq \sum_A x_A \cdot A \cap \{v_0, v_1\} $; no acyclic set contains both digon ends, so the sum is $\leq W \leq 2$; the equality chain forces $ A \cap \{v_0, v_1\} = 1$ on every support set (and $W = 2$ exactly). Uses D nonempty, which the writeup assumes.

step	label	note
7. Forward implication ($\chi_{\vec{f}}(R(D)) \leq 2 \implies \chi(\vec{D}) \leq 2$)	VALID	Any support set A is a digon transversal, so C_0, C_1 partition $V(D)$; within A there are no cross-layer arcs at all (the only cross-layer arcs are digon arcs, and A has one end of each), so $H[A]$ is the disjoint union of induced copies of $D[C_0]$ and $D[C_1]$; acyclicity of $H[A]$ gives an acyclic 2-colouring of D .
8. Reverse implication ($\chi(\vec{D}) \leq 2 \implies \chi_{\vec{f}}(R(D)) \leq 2$)	VALID	$A = C_0 \times \{0\} \cup C_1 \times \{1\}$ and its complement B each contain exactly one end of every digon and induce disjoint unions of $D[C_0]$ and $D[C_1]$ copies, hence are acyclic; weights $1 + 1$ cover every vertex once with total 2.
9. Complexity conclusion	VALID	Polynomial many-one reduction + NP membership $\implies$ NP-completeness of FDC(2); together with the paper's Theorem 3.20 this covers all $p > 1$.
10. Scope remark (digons essential)	VALID	Accurate and honest: the oriented-graph restriction is not settled, but Problem 3.21 does not ask for it.

No GAP or ERROR found. Every inequality was re-derived by hand and the combinatorial claims verified by machine (below).

B.3 Reference check

- **Hochstättler, Schröder, Steiner, arXiv:1812.02420v4 / DMTCS 22:1 (2020).** Downloaded the full LaTeX source from arXiv. Confirmed: Problem 3.21 (“Let $p \geq 1$ be a fixed real number. Instance: A (multi-)digraph D . Decide whether $\chi_{\vec{f}}(D) \leq p$.”); the LP definition of $\chi_{\vec{f}}$; the digon/anti-parallel-arc conventions; NP membership for all $p \geq 1$ (Observation 3.15); NP-completeness for $p > 2$ (Observation 3.16, via $\chi_{\vec{f}}$ of undirected graphs) and for $p \in (1,2)$ (via l-splits); and the concluding remark explicitly leaving $p = 2$ open while predicting NP-completeness. All facts the writeup imports from the source paper are used with their exact hypotheses.
- **Bokal, Fijavž, Juvan, Kayll, Mohar (2004), “The circular chromatic number of a digraph”, JGT 46(3):227–240.** Confirmed to prove NP-completeness of deciding $\chi(\vec{D}) \leq 2$ for digraphs — the sole external hardness result the reduction needs. Also quoted in the source paper’s abstract as its premise.
- **Novelty check.** Searched for any post-2020 resolution of the $p = 2$ case (“fractional dichromatic number” + $p = 2$ / NP-complete / complexity, papers citing 1812.02420). Found none; the catalog’s own auto-review (May 2026, web-search enabled) also found none. I cannot prove a negative, but I found no evidence this is ALREADY_KNOWN.

B.4 Computational check

Script: `verification/scripts/1812.02420__02/check_reduction.py` (Python 3, networkx + scipy/HiGHS; brute-force enumeration of all acyclic vertex subsets of $R(D)$ — a MultiDiGraph, so digons register as directed 2-cycles — and exact solution of the covering LP; $\chi^{\rightarrow}(D) \leq 2$ checked by brute force over all 2-partitions).

Results (all 125 checks passed):

- Directed triangle C_3 ($\chi^{\rightarrow} = 2$): $\chi^{\rightarrow}_f(R(C_3)) = 2.000000$. Equivalence holds.
- Bidirected K_3 ($\chi^{\rightarrow} = 3$): $\chi^{\rightarrow}_f(R(K_{\leftrightarrow 3})) = 3.000000 > 2$. Equivalence holds.
- Single digon ($\chi^{\rightarrow} = 2$): $\chi^{\rightarrow}_f(R) = 2.000000$.
- **Paley tournament ST_7** (digon-free, $\chi^{\rightarrow} = 3$ — the smallest 3-dichromatic tournament, a nontrivial stress test since $\chi^{\rightarrow}_f$ of ST_7 itself is below 2): $\chi^{\rightarrow}_f(R(ST_7)) = 2.333333 = 7/3 > 2$. Equivalence holds. (14 vertices, all 2^{14} subsets enumerated.)
- Single vertex: $\chi^{\rightarrow}_f(R) = 2.000000$ (the forcing lower bound $\chi^{\rightarrow}_f(R(D)) \geq 2$ for nonempty D verified throughout).
- Sanity: $\chi^{\rightarrow}_f(C_3) = 1.500000 = 3/2$ as expected, confirming the LP encoding.
- 120 random digraphs, $n \in [1, 6]$, densities 0.2–0.7 (many containing digons): the equivalence $\chi^{\rightarrow}_f(R(D)) \leq 2 \iff \chi^{\rightarrow}(D) \leq 2$ held in every trial (observed values: exactly 2 when $\chi^{\rightarrow}(D) \leq 2$, and $\geq 7/3$ when $\chi^{\rightarrow}(D) > 2$), and in every yes-instance every optimal LP support set was a digon transversal, exactly as the forcing lemma predicts.

No computational discrepancy of any kind was found.

B.5 Caveats

- **Nonempty instance.** The forcing lemma needs $V(D) \neq \emptyset$; the writeup assumes D nonempty. The empty digraph is a trivial yes-instance on both sides, so the reduction extends with a one-line remark. Cosmetic.
- **Loops.** The paper declares digraphs loopless, so the LP is always feasible; no issue.
- **Attainment of the LP optimum.** The argument tacitly uses that $\chi^{\rightarrow}_f \leq 2$ yields an actual feasible solution of weight ≤ 2 ; true because $\chi^{\rightarrow}_f$ is a finite LP minimum (attained), consistent with the paper’s own NP-membership proof. Not a gap.
- **Digons are essential** (flagged by the writeup itself): the complexity of deciding $\chi^{\rightarrow}_f \leq 2$ for *oriented* (digon-free) digraphs remains open. Problem 3.21 explicitly allows multidigraphs, so this does not affect the verdict on the catalogued problem.
- **Surprising simplicity.** The reduction is elementary enough that its being missed for six years is mildly suspicious; this motivated the exhaustive hand re-derivation, the source-fetch of the paper, the literature search for prior resolution (none found), and the machine verification, all of which the argument survived. Residual risk is essentially limited to an unindexed prior appearance of the same trick (which would change the verdict to `ALREADY_KNOWN`, not to incorrect).
- **“Proved” semantics.** The catalog problem is a complexity question, not a yes/no conjecture; “proved” here means the open case $p = 2$ is resolved (as NP-complete, the outcome the authors predicted). The writeup’s verdict block says exactly this.

B.6 Referee summary

I set out assuming the writeup was wrong and failed to break it. The interpretation is exactly the paper’s: Problem 3.21 takes multidigraphs (the source explicitly allows anti-parallel arcs and calls a digon a directed 2-cycle) and $\chi^{\rightarrow}_f$ is the acyclic-set covering LP, verbatim. The reduction — two disjoint copies of D joined by a perfect matching of digons — is checked step by step: the digon coverage/packing argument rigidly forces every support set of a weight-2 fractional colouring to be a layer transversal, whose two sides induce copies of D on complementary vertex sets, giving $\chi^{\rightarrow}_f(R(D))$

$\leq 2 \iff \chi(\vec{D}) \leq 2$; combined with Bokal et al.'s 2004 NP-completeness of 2-dichromatic colourability (confirmed) and the source paper's own NP membership observation, NP-completeness of the $p = 2$ case follows. Independent brute-force LP computation confirmed the equivalence on 125 instances, including the digon-free Paley tournament ($\chi_{\text{f}}(\vec{R}(\text{ST}_7)) = 7/3$) and bidirected K_3 ($\chi_{\text{f}} = 3$). The paper's own techniques provably could not reach $p = 2$ (one covers only $p > 2$, the other only $p \in (1,2)$), and its conclusion explicitly solicits an NP-completeness proof "using other techniques," which this is. No prior resolution was found in the literature. Verdict: CONFIRMED.

References

- [1] D. Bokal, G. Fijavž, M. Juvan, P. M. Kayll and B. Mohar, The circular chromatic number of a digraph, *J. Graph Theory* 46 (2004) 227–240.
- [2] W. Hochstättler, F. Schröder and R. Steiner, On the complexity of digraph colourings and vertex arboricity, *Discrete Math. Theor. Comput. Sci.* 22:1 (2020); arXiv:1812.02420.